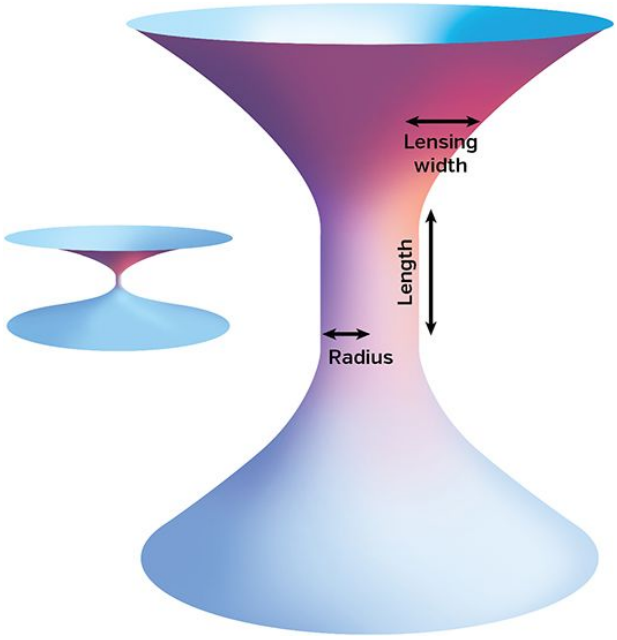


Fig. 15.1. A wormhole viewed from the bulk and my three handles for adjusting its shape. (The inset on the left is the same wormhole, viewed from farther away in the bulk so we see its outer parts.)

The first handle is the wormhole's *radius* as measured by an ultra-advanced engineer looking in from the bulk (analog of Gargantua's radius). If we multiply that radius by $2\pi = 6.28318...$, we get the wormhole's circumference as measured by Cooper when he pilots the *Endurance* around or through it. Chris chose the radius before I began to work. He wanted the wormhole's gravitational lensing of stars to be barely visible from Earth with the best large-telescope technology then available to NASA. That fixed the radius at about a kilometer.

The second handle is the wormhole's *length*, as measured equally well by Cooper or by an engineer in the bulk.

The third handle determines how strongly the wormhole lenses the light from objects behind it. The details of the lensing are fixed by the shape of space near the wormhole's mouths. I chose that shape similar to the shape of space outside the horizon of a nonspinning black hole.